Kevin Pugliese

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EDUCATION

Bachelor of Science in Computer Engineering *December 2027*
University of Florida Honors Program | Gainesville, FL *GPA: 3.67*

Relevant Coursework: *Digital Logic & Computer Systems, Computer Organization, Computational Linear Algebra, Data Structures and Algorithms, Operating Systems*
Fall 2026: *Digital Design, Microprocessor Applications, Signals & Systems*
Spring 2027: *Parallel Computer Architecture, CAD for Hardware Security Validation*

EXPERIENCE

Hardware-Software Integration Specialist Co-op *May 2026 - August 2026*
CAE USA, Tampa, Florida

- Replaced an unmaintained 10-year-old C# node inventory tool with a single PowerShell script, cutting a 30-minute manual process required on every shipped simulator to under one minute
- Automated Juniper switch configuration generation by extending an internal Jinja/XML tool; validated output against previously hand-maintained configs
- Traveled to an active Air Force base to support program deployment, performing daily simulator bring-up for customer testing and modifying cockpits to match aircraft hardware changes

Technical Specialist / AI Data Annotator *December 2025 - Present*
Handshake AI Solutions, Remote

- Grade coding agent communication against a written rubric; promoted to reviewer, auditing other annotators' work and flagging LLM-generated submissions
- Design multistep tasks in KiCad and Defold using proprietary model evaluation tools

PROJECTS

FPGA AM Radio Receiver | *SystemVerilog, C, Python, Quartus, QuestaSim* *November 2026*

- Designed a 16-channel AM radio receiver on an Intel MAX 10 FPGA that samples at 64 MHz
- Isolated each station with I/Q mixing and CIC/FIR decimation filters, cutting 64M samples/s to 16k
- Reused one I and one Q multiplier for all 128 FIR taps of all 16 channels, achieving 81.3 MHz Fmax (64 MHz clk) with 37,624 LEs
- Linked the FPGA to an ESP32 over SPI and wrote C firmware to stream audio and tune over Wi-Fi

Elementary CPU | *VHDL, Quartus* *April 2026*

- Designed a 4-bit CPU with ROM-based instruction fetch and a registered ALU, achieving 250 MHz Fmax in 57 logic elements and 17 registers on an Intel MAX 10 FPGA
- Implemented the control unit in VHDL as a state machine decoding 6 opcodes into datapath control words, including multi-state instructions for data loads and jumps
- Verified all instructions in simulation and on hardware using hand-assembled test programs

tModLoader | *C#, GitHub* *October 2025*

- Authored and merged a PR to the official tModLoader repository (10M+ downloads)
- Independently fixed a C# rendering bug that caused some objects to appear misaligned

GraphUF | *Flask, Git, Nginx, Python, Bash* *July 2025 - August 2025*

- Supplemented UF course catalog with Flask web app to explore transitive course unlocks
- Collaborated with two teammates, handling Git merge conflicts, pull requests, and code reviews
- Processed 8M+ unique data points by scraping and cleaning UF Courses API
- Created transitive closure map of 80k-node DAG to achieve O(1) retrieval of reachable nodes

CERTIFICATIONS

GIAC Foundational Cybersecurity Technologies *August 2023 - August 2027*

SKILLS

Languages: SystemVerilog, VHDL, Assembly, C, C++, Python, C#, MATLAB, Java, PowerShell, Bash
Tools: Quartus, QuestaSim, Git, Linux, Jira, Docker, LTspice, KiCad, Fusion 360